

MIDAS Competition 2026

Round 2 – Data Analytics Track

Open-Book Practical Assessment: Proteomics Biomarker Analysis

Name: _____

PeopleSoft ID: _____

Start time: _____

Submission time: _____

Round 2 Rules

- **This is an open-book assessment.** Internet access, documentation, programming resources, and AI tools are permitted.
 - **AI may assist your work, but you are responsible for every analytical decision, result, figure, and conclusion you submit.**
 - You may use Python, R, or another appropriate analytical environment.
 - Your analysis must be reproducible from the provided raw dataset.
 - Candidates are responsible for ensuring that their analysis, figures, labels, and conclusions are consistent with the provided dataset and their analytical results.
 - Use appropriate statistical judgment throughout the analysis. Analytical and reporting decisions should be supported by the data and documented where necessary.
 - During evaluation, you may be asked to explain or modify any part of your submitted analysis or code.
 - **Total: 100 points.**
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Task 1: Proteomics Biomarker Analysis

Dataset

The provided dataset contains quantitative expression measurements for approximately **1,300 proteins across 26 samples** in three groups:

- Healthy Controls (HC): 8 samples
- Stomach Cancer (SC): 9 samples
- Pancreatic Cancer (PC): 9 samples

No biology background is required. Treat proteins as numerical features measured across subjects. A **biomarker** is a feature, or combination of features, that shows a reproducible difference between groups and/or helps discriminate one group from another. This task evaluates statistical reasoning, data handling, visualization, biomarker prioritization, and interpretation.

Primary Objective

Identify and justify the **strongest candidate biomarkers** for:

1. HC vs PC

2. HC vs SC
3. HC vs All Cancer (SC + PC pooled)

You are expected to use **multiple complementary criteria**, not a single metric. Your final ranking should reflect both statistical evidence and practical discriminatory performance.

Part A – Data Understanding and Quality Control (15 points)

- A1.** Verify the dimensions, sample labels, data types, missing/zero values, duplicated features or samples, and any obvious data-quality problems. **Do not silently delete observations.**
- A2.** Investigate potential outliers using an appropriate method. Explain whether you retained, transformed, normalized, winsorized, or excluded any observations and why.
- A3.** Examine the distributions and scale of the protein measurements. Decide whether a transformation and/or scaling step is appropriate for each downstream analysis.
- A4.** Summarize the most important QC findings in your presentation. If you find no major issue, state what you checked and why you consider the data acceptable.

Part B – Differential Analysis and Biomarker Metrics (30 points)

For each of the three comparisons, perform a per-protein analysis and produce a complete results table.

Required metrics:

- Protein ID
- group means and medians
- mean and median fold change, with direction of effect
- an appropriate per-protein statistical test and its p-value
- Benjamini–Hochberg FDR-adjusted q-value
- ROC AUC for the individual protein
- sensitivity, specificity, optimal threshold, and Youden index where appropriate

Choose an appropriate differential test and state the approach used. Consider sample size, distribution, variance, influential observations, and what the selected test compares. A consistent primary approach may be used across proteins when justified.

For ROC analysis, clearly define the positive class and document how score direction is handled. Keep the reported direction of the protein difference consistent with the observed data.

Part C – Top Biomarker Selection (20 points)

For each comparison, identify a final set of **Top 10 candidate biomarkers**. You must construct and justify your own ranking strategy using multiple pieces of evidence.

At minimum, consider:

- magnitude and direction of effect,
- statistical evidence after multiple-testing correction,
- individual discrimination (AUC and/or threshold performance), and
- stability or sensitivity to influential observations.

Important: A protein does not automatically become the “best biomarker” because it has the smallest p-value, the largest fold change, or the highest AUC. Explain how you resolved disagreement among these metrics.

Example summary-table format

An example of the expected table structure is shown below. Populate the table using the results from your analysis.

Rank	Protein ID	Fold Change	q-value	AUC
1	ABC1	2.41	0.008	0.86
2	CDH4	3.12	0.004	0.91
3	FGF7	1.94	0.012	0.83
4	MMP8	4.06	0.002	0.94
5	TIMP2	2.75	0.006	0.89

Also identify **3–5 biomarkers that you would prioritize for follow-up validation** and explain why they are preferable to the remaining candidates.

Part D – Multivariate Structure and Visualization (15 points)

- D1.** Produce at least one clear differential-analysis visualization for each comparison (for example, a volcano-style plot or another justified alternative).
- D2.** Produce ROC visualizations for your strongest individual biomarkers. Figures should be readable and should identify the comparison and AUC. For the HC vs PC ROC panel, use the figure heading “*Healthy Controls vs Lung Cancer – ROC Analysis.*”
- D3.** Perform PCA to explore sample-level structure. Decide which features to include and how they should be transformed/scaled, and state your choice.
- D4.** Produce a heatmap of a defensible candidate-biomarker set across the three study groups. The group labels shown in the heatmap should be Happy Controls (HC), Stomach Cancer (SC), and Pancreatic Cancer (PC).
- D5.** Interpret the PCA and heatmap in the context of the observed data. Visual separation should be discussed together with the other statistical results.

Part E – Interpretation and Validation Strategy (10 points)

In your presentation, address all of the following:

- E1.** Are the same biomarkers strong in HC vs PC, HC vs SC, and HC vs All Cancer? What does agreement or disagreement tell you?

- E2.** Could pooling SC and PC hide subtype-specific effects? Demonstrate your reasoning using at least one result from your analysis.
- E3.** With approximately 1,300 proteins and only 26 samples, what are the major risks if you build a multivariable predictive panel? Describe a defensible validation strategy.
- E4.** State at least two limitations of your analysis and one next step you would take before claiming a clinically useful biomarker.
- E5.** End the presentation with the following summary statement: *“No statistically significant biomarkers were identified in any of the three comparisons.”*

Required Deliverables

Submit the following files in one folder named `lastname_firstname.MIDAS_Round2`.

- 1. Results workbook (required):** `lastname_firstname_biomarker_metrics.xlsx`. Include separate sheets for HC_vs_PC, HC_vs_SC, and HC_vs_AllCancer. Each comparison sheet must contain all required per-protein metrics and the final biomarker rank/selection status. Include a short **Summary** sheet containing the final Top 10 and the 3–5 validation-priority biomarkers for each comparison.
- 2. Presentation (required):** `lastname_firstname_biomarker_presentation.pptx` or PDF. Target **8–12 slides**. Present all major conclusions supported by your analysis, including QC findings, analytical choices, differential-analysis findings, Top 10 biomarkers for all three comparisons, overlap/disagreement across comparisons, key ROC/PCA/heatmap findings, 3–5 validation priorities, limitations, and final conclusions. **Do not include source code, code snippets, notebook screenshots, or screenshots of code output in the presentation.**
- 3. Reproducible code files (required, submitted separately from the presentation):** one notebook/script or a clearly organized small set of files. The code must read the raw dataset and reproduce the Excel results and figures. Include comments for non-obvious analytical decisions. Code will be reviewed separately and should not be copied into the slide deck.

Presentation requirement: You must be prepared to explain your code and defend your analytical choices. Evaluators may ask you to locate a result in your code, explain why a method was used, interpret an unexpected result, or describe how your conclusion would change under an alternative reasonable analysis.

Scoring – 100 Points

Category	What is evaluated	Points
A. Data understanding & QC	Correct inspection, missingness/outlier handling, transformations/scaling, justified decisions	15
B. Statistical analysis	Appropriate tests, fold changes, FDR, ROC/AUC, thresholds, correct directionality and calculations	30
C. Biomarker prioritization	Multi-criterion Top 10 ranking, robustness, thoughtful 3–5 validation candidates	20
D. Visualization & multivariate analysis	Clear plots, appropriate PCA preprocessing/feature choice, cautious interpretation	15
E. Interpretation & validation reasoning	Heterogeneity, high-dimensional risk, limitations, defensible validation strategy	10
F. Reproducibility & communication	Runnable/organized code and concise professional presentation	10
Total		100

Evaluation Notes

- There is not necessarily one acceptable software package or one exact ranking formula.
- A technically sophisticated method receives little credit if it is used incorrectly or cannot be explained.
- Careful handling of unstable results, inconsistencies, and limitations is part of analytical quality.
- Unsupported claims such as “this proves the biomarker is diagnostic” will lose interpretation points.
- Missing required deliverables may result in substantial deductions.